

forecasting is hard, clues to better ground us in the current reality can be found by looking at the current term structure of interest rates and stock market valuations.

More generally, we have little basis for knowing how suitable our simulation input assumptions truly are. Will our assumptions be representative of future financial market returns? Small changes in assumptions for portfolio returns and volatility can create a large impact on sustainable retirement spending, as efforts to test assumptions with the PMT and NPER formulas clearly show.

The International Experience of the 4 Percent Rule

An argument in support of the 4 percent rule is that the post-1926 US historical period included a number of calamitous market events (Great Depression, Great Stagnation of the 1970s, etc.). As such, the argument goes, it is hard to imagine an even more dire situation awaiting future retirees. The historical success of the 4 percent rule suggests that we can reasonably plan for its continued success in the future.

But from a global perspective, asset returns enjoyed a particularly favorable climate in the twentieth-century United States, and to the extent that the United States may experience a more typical outcome in the twenty-first century, present conceptions of safe withdrawal rates may be less safe. Prospective retirees must consider whether they are comfortable basing retirement decisions on the impressive but perhaps anomalous numbers found in historical US data. In planning for retirements in the future, it is unclear whether asset returns of this century will continue to be as favorable as they were in the twentieth century, or whether savers and retirees should plan for something closer to the average international experience.

My first foray into researching personal retirement planning—“An International Perspective on Safe Withdrawal Rates: The Demise of the 4% Rule?” (published in the December 2010 *Journal of Financial Planning*)—was a study of the sustainability of the 4 percent rule in other developed-market countries. The study is based on a data set providing financial market returns since 1900 for twenty developed-market countries plus GDP-weighted world and world ex-US indexes. The data is from the Dimson-Marsh-Staunton Global Returns Dataset provided by Ibbotson and Morningstar.

From an international perspective, a 4 percent withdrawal rate has been problematic. In the original article, to avoid claims of bias that I chose an asset